A system of nomenclature known as "Device Function Numbering" has been adopted by the Institute of Electrical and Electronic Engineers (IEEE) and the American National Standards Institute (ANSI). It provides a means of quickly grasping the main purpose of any device used as a part of an automatic and manual station control, supervisory, and associated telemetering equipment. This system, first developed in connection with automatic switchgear, is now applied to all types of control, protection and power distribution equipment. It is based on the assignment of a standard number to each of the several fundamental functions performed by the component elements of a complete equipment.

These device functions may refer to the actual function the device performs or may refer to the electrical or application to which the device is responsive. Therefore, there may be in some instances a choice of the function number used for a given device. The best choice is the one which is recognized to have the narrowest interpretation so that it most specifically identifies the device in the minds of everyone involved in the design, start-up, operation and maintenance of the equipment.

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The device function numbers with appropriate suffix letter or letters, where necessary, are used on electrical diagrams, in instruction books and in specifications. Device function numbers, each with its corresponding function name and the general description of each function, are listed in this article.

When alternate names and descriptions are included under the function, only the name and description which applies to each specific case should be used. In general, only one name for each device, such as relay, contactor, circuit breaker, switch, or device, is included in each function designation. However, when the function is not restricted to any specific type of device and where the type of device itself is merely incidental, any one of these alternative names may be substituted. For example, if a contactor is used in place of a device function 6, starting circuit breaker, the function name should be specified as Starting Contactor.

Device Number	Device Name	Functional Description	Common GE Application
1	Master Element	Initiating device, such as a control switch, etc. which serves either directly or through such permissive devices as protective and time-delay relays to place equipment in or out of operation. **Note:** This number is normally used for a hand-operated device, although it may also be used for an electrical or mechanical device for which no other function number is suitable.	Master control switch/relay, e.g., automatic start/stop.
2	Tme-Delay Starting or Closing Relay	Device that gives a desired amount of time delay before or after any point of operation 'in a switching sequence or protective relay system, except as specifically provided by device functions 48, 62, and 79.	Sequence timer, e.g. firing timer, fire protection CO2 discharge timer, etc.
3	Checking or Inter-Locking Relay	Relay that operates in response to the position of a number of other devices (or to a number of predetermined conditions) in an equipment, to allow an operating sequence to proceed or to stop, or to provide a check of the position of these devices or of these conditions for any purpose.	Checking relay, e.g., sequence completion, shaft voltage monitor/alarm, etc.
4	Master Contactor	Device, generally controlled by device function 1 or the equivalent and the required permissive and protective devices, that serves to make an d break the necessary control circuits to place an equipment into operation under the desired conditions and to take it out of operation under abnormal conditions.	Mast control relay, e.g., starting motor, fuel forwarding, etc.
5	Stopping Device	Control device used primarily to shut down an equipment and hold it out of operation. (This device may be manually or electrically actuated, but excludes the function of electrical lockout [see device function 86] on abnormal conditions.)	Trip Switch, e.g., emergency trip, pushbutton station, etc.
6	Starting Circuit Breaker	Device whose principal function is to connect a machine to its source of starting voltage.	Circuit breaker, e.g., battery charger, etc.
7	Rate-of-Rise Relay	Relay that functions on an excessive rate-of-rise of current.	Anode circuit breaker.
8	Control Power Disconnecting Device	Disconnecting device, such as a knife switch, circuit breaker, or pull-out fuse block, used for the purpose of respectively connecting and disconnecting the source of control power to and from the control bus or equipment. **Note:** Control power is considered to include auxiliary power which supplies such apparatus as small motors and heaters.	Power transfer mechanism.

Device Number	Device Name	Functional Description	Common GE Application
9	Reversing Device	Device that is used for the purpose of reversing a machine field or for performing any other reversing functions.	
10	Unit Sequence Switch	Switch that is used to change the sequence in which units may be placed in and out of service in multiple-unit equipment.	
11	Multi-Function Device	Device that performs three or more comparatively important functions which could only be designated by combining several of these device function numbers. All of the functions performed by device 11 shall be defined in the drawing legend or device function definition list. **Note:** If only two relatively important functions are performed by the device, it is preferred that both function numbers be used, as described in 3.6.	
12	Overspeed Device	Usually a direct-connected Turbine overspeed trip speed switch which functions on machine overspeed.	Turbine Overspeed Trip.
13	Synchronous Speed Device	Device such as a centrifugal-speed switch, a slip-frequency relay, a voltage relay, an undercurrent relay, or any type of device that operates at approximately the synchronous speed of a machine.	
14	Underspeed Device	Device that functions when the speed of a machine falls below a predetermined value.	Speed relay, i.e., pick-up/drop-out @ specific frequency/RPM.
15	Speed or Frequency Matching Device	Device that functions to match and hold the speed or frequency of a machine or of a system equal to, or approximately equal to, that of another machine, source or system.	Speed matching relay.
16		Reserved for Future Application	
17	Shunting or Discharge Switch	Switch that serves to open or to close a shunting circuit around any piece of apparatus (except a resistor), such as a machine field, a machine armature, a capacitor, or a reactor. **Note:** This excludes devices that perform shunting operations as may be necessary in starting a machine by devices 6 or 42 (or equivalent), and excludes device function 73 that serves for the switching of resistors.	

Device Number	Device Name	Functional Description	Common GE Application
18	Accelerating or Decelerating Device	Device that is used to close or to cause the closing of circuits which are used to increase or decrease the speed of a machine.	Starting motor accelerating contactor.
19	Starting-to-Running Transition Contactor	Device that operates to initiate or cause the automatic transfer of a machine from the starting to the running power connection.	
20	Electrically Operated Valve	This is an electrically operated, controlled, or monitored valve used in a fluid, air, gas or vacuum line. **Note:** The function of the valve may be more completely indicated by the use of suffixes as discussed in 3.2.	
21	Distance Relay	Relay that functions when the circuit admittance, impedance, or reactance increases or decreases beyond a predetermined value.	
22	Equalizer Circuit Breaker	Breaker that serves to control or to make and break the equalizer or the current-balancing connections for a machine field, or for regulating equipment, in a multiple-unit installation.	
23	Temperature Control Device	Device that functions to raise or lower the temperature of a machine or other apparatus or of any medium, when its temperature falls below, or rises above, a predetermined value. **Note:** An example is a thermostat that switches on a space heater in a switch-gear assembly when the temperature falls to a desired value. This should be distinguished from a device that is used to provide automatic temperature regulation between close limits and would be designated as device function 90T.	Heaters, e.g., lube oil, compartment space heater, etc.
24	Volts per Hertz Relay	Relay that functions when the ratio or voltage to frequency exceeds a present value. The relay may have an instantaneous or a time characteristic.	
25	Synchronizing or Synchronism Check Device	Device that operates when two ac circuits are within the desired limits of frequency, phase angle, and voltage, to permit or cause the paralleling of these two circuits.	Synchronizing relay.

Device Number	Device Name	Functional Description	Common GE Application
26	Apparatus Thermal Device	Device that functions when the temperature of the protected apparatus (other than the load-carrying windings of machines and transformers as covered by device function number 49), or of a liquid or other medium exceeds a predetermined value; or when the temperature of the protected apparatus or of any medium decreases below a predetermined value.	Temperature switch, e.g., lube oil heater permissive.
27	Undervoltage Relay	Relay which operates when its input voltage is less than a predetermined value.	Undervoltage relay, e.g., bus, MCC, generator H2 cabinet, etc.
28	Flame Detector	Device that monitors the presence of the pilot or main flame in such apparatus as a gas turbine or a steam boiler.	Flame scanner.
29	Isolating Contactor	Device that is used expressly for disconnecting one circuit from another for the purposes of emergency operation, maintenance, or test.	
30	Annunciator Relay	Non-automatically reset device that gives a number of separate visual indications upon the functioning of protective devices, and which may also be arranged to perform a lockout function.	Annunciator relay.
31	Separate Excitation Device	Device that connects a circuit, such as the shunt field of a synchronous converter, to a source of separate excitation during the starting sequence.	
32	Directional Power Relay	Relay which operates on a predetermined value of a power flow in a given direction, or upon reverse power flow such as that resulting from the motoring of a generator upon loss of its prime mover.	Directional power relay.
33	Position Switch Switch	Switch that makes or breaks contact when the main device or piece of apparatus which has no device function number reaches a given position	Limit switch, e.g., vent damper position switch, valve position switch, etc.

Device Number	Device Name	Functional Description	Common GE Application
34	Master Sequence Device	Device such as a motor operated multicontact switch, or the equivalent, or a programming device, such as a computer, that establishes or determines the operating sequence of the major devices in an equipment during starting and stopping or during other sequential switching operations.	Master sequence device, e.g., air filter self-cleaning sequencer.
35	Brush Operating or Slip-Ring Short-Circulating Device	Device for raising, lowering, or shifting the brushes of a machine, or for short- circuiting its slip rings, or for engaging or disengaging the contacts of a mechanical rectifier.	
36	Polarity or Polarizing Voltage Device	Device that operates, or permits the operation of, another device on a predetermined polarity only, or verifies the presence of a polarizing voltage in an equipment.	
37	Undercurrent or Underpower Relay	Relay that functions when the current or power flow decreases below a predetermined value.	Undercurrent or underpower relay, e.g., low battery charger output current.
38	Bearing Protective Device	Device that functions on excessive bearing temperature, or on other abnormal mechanical conditions associated with the bearing, such as undue wear, which may eventually result in excessive bearing temperature or failure.	Bearing protective device, e.g., high drain temperature alarm.
39	Mechanical Condition Monitor	Device that functions upon the occurrence of an abnormal mechanical condition (except that associated with bearings as covered under device function 38), such as excessive vibration, eccentricity, expansion, shock, tilting, or seal failure	Vibration sensor/switch, trip relay.
40	Field Relay	Relay that functions on a given or abnormally low value or failure of machine field current, or on excessive value of the reactive component of armature current in an ac machine indicating abnormally low field excitation.	Loss of excitation relay.
41	Field Circuit Breaker	Device that functions to apply or remove the field excitation of a machine.	Field flashing suppression contactor.

Device Number	Device Name	Functional Description	Common GE Application
42	Running Circuit Breaker	Device whose principal function is to connect a machine to its source of running or operating voltage. This function may also be used for a device, such as a contactor, that is used in series with a circuit breaker or other fault protecting means, primarily for frequent opening and closing of the circuit.	
43	Manual Transfer or Selector Device	Manually-operated device that transfers the control circuits in order to modify the plan of operation of the switching equipment or of some other the devices.	Master operation selector switch.
44	Unit Sequence Starting Relay	Relay that functions to start the next available unit in a multiple-unit equipment upon the failure or non-availability of the normally preceding unit.	
45	Atmosphere Condition Monitor	Device that functions upon the occurrence of an abnormal atmospheric condition, such as damaging fumes, explosive mixtures, smoke, or fire.	Fire detector/ protection actuation switch.
46	Reverse-Phase or Phase-Balance Current Relay	Relay that functions when the polyphase currents are of reverse-phase sequence, or when the polyphase currents are unbalanced or contain negative phase-sequence components above a given amount.	Negative sequence relay.
47	Phase-Sequencer Phase-Balance Voltage Relay	Relay that functions upon a predetermined value of polyphase voltage in the desired phase-sequence, or when the polyphase voltages are unbalanced, or when the negative phase-sequence voltage exceeds a given amount.	Phase sequence relay.
48	Incomplete Sequence Relay	Relay that generally returns the equipment to the normal, or off, position and locks it out if the normal starting, operating, or stopping sequence is not properly completed within a predetermined time.	Incomplete Sequence Relay, e.g., hydraulic ratchet cycling sequence timer.
49	Machine or Transformer Thermal Relay	Relay that functions when the temperature of a machine armature winding or other load-carrying winding or element of a machine or power transformer exceeds a predetermined value.	Motor overload protection.
50	Instantaneous Overcurrent	Relay that functions instantaneously on an excessive value of current.	Instantaneous phase/ground fault relay, over/under current relay.

Device Number	Device Name	Functional Description	Common GE Application
51	AC Time Overcurrent Relay	Relay that functions when the ac input current exceeds a predetermined value, and in which the input current and operating time are inversely related through a substantial portion of the performance range.	System phase/ground fault relay.
52	AC Circuit Breaker	Device that is used to close and interrupt an ac power circuit under normal conditions or to interrupt this circuit under fault or emergency conditions	AC load switchgear/ contactor, e.g., motor heater, etc.
53	Exciter or DC Generator Relay	Relay that forces the dc machine field excitation to build up during starting or which functions when the machine voltage has built up to a given value.	
54	Turning Gear Engaging Device	Device is an electrically-operated, controlled, or monitored device which functions to cause the turning gear to engage (or disengage) the machine shaft.	
55	Power Factor Relay	Relay that operates when the power factor in an ac circuit rises above or falls below a predetermined value.	
56	Field Application Relay	Relay that automatically controls the application of the field excitation to an ac motor at some predetermined point in the slip cycle.	
57	Short-Circuiting or Grounding Device	Primary circuit switching device that functions to short circuit or to ground a circuit in response to automatic or manual means.	
58	Rectification Failure Relay	Device that functions if a power rectified fails to conduct or block properly.	
59	Over-Voltage Relay	Relay which operates when its input voltage is more than a predetermined value	Over-voltage relay, voltage regulator, generator volts/hertz ratio.
60	Voltage or Current Balance Relay	Relay that operates on a given difference in voltage, or current input or output of two circuits.	Current/voltage balance relay.
61	Density Switch or Sensor	Device which operates on a given value, or a given rate of change of gas density.	

Device Number	Device Name	Functional Description	Common GE Application
62	Time-Delay Stopping or Opening Relay	Time-delay relay that serves in conjunction with device that initiates the shutdown, stopping, or opening operation in an automatic sequence or protective relay system.	Timers, e.g., coastdown, cooldown, fuel purge, etc.
63	Pressure Switch	Switch which operates on given values, or on a given rate of change of pressure.	Pressure switch, e.g., lube oil pressure, fuel supply pressure, etc.
64	Ground Detector Relay	Relay that operates on failure of machine or other apparatus insulation to ground. **Note:** This function control circuit, etc. is not applied to a device connected in the secondary circuit of current transformers in a normally grounded power system, where other device numbers with a suffix G or N should be used, that is, 5 IN for an ac time over-current delay connected in the secondary neutral of the current transformers.	Ground detector/protective relay, e.g., time, generator field.
65	Governor	Assembly of fluid, electrical, or mechanical control equipment used for regulating the flow of water, steam, or other medium to the prime mover for such purposes as starting, holding speed or load, or stopping.	Speed governor, e.g., diesel starting means, fuel regulator, turbine nozzle regulator, servo valve, etc.
66	Notching or Jogging Device	Device that functions to allow only a specified number of operations of a given device, or equipment, or a specified number of successive operations within a given time of each other. It is also a device that functions to energize a circuit periodically or for factions of specified time intervals, or that is used to permit intermittent acceleration of jogging of a machine at low speeds for mechanical positioning.	Filter media measuring switch.
67	AC Directional Overcurrent Relay	Relay that functions on a desired value of ac overcurrent flowing in a predetermined direction.	VAR control relay.

Device Number	Device Name	Functional Description	Common GE Application
68	Blocking Relay	Relay that initiates a pilot signal for blocking of tripping on external faults in a transmission line or in other apparatus under predetermined conditions, or cooperates with other devices to block tripping or to block reclosing on an out-of-step condition or on power swings.	Reclosing block relay.
69	Permissive Control Device	Generally a two-position device that in one position permits the closing of a circuit breaker, or the placing of an equipment into operation, and in the other position prevents the circuit breaker or the equipment from being operated.	Turbine/ compressor washing system permissive switch.
70	Rheostat	Variable resistance device used in an electric circuit, which is electrically operated or has other electrical accessories, such as auxiliary, position, or limit switches.	Motor-operated rheostat, e.g., turbine nozzle regulation; mechanically drive position indicator, e.g., inlet guide vanes.
71	Level Switch	Switch which operates on given values, or on a given rate of change of level	Liquid or gas level relay, e.g., lube oil, CO2, etc.
72	DC Circuit Breaker	Circuit breaker used to close and interrupt a dc power circuit under normal conditions or to interrupt this circuit under fault or emergency conditions.	DC load switchgear/contact or, e.g., emer. motors.
73	Load-Resistor Contactor	Contactor used to shunt or insert a step of load limiting, shifting, or indicating resistance in a power circuit, or to switch a space heater in circuit, or to switch a light or regenerative load resistor of a power rectifier or other machine in and out of circuit.	Seal oil pump overload relay.
74	Alarm Relay	Relay other than an annunciator, as covered under device function 30, that is used to operate, or to operate in connection with, a visual or audible alarm.	Seal oil pump overload relay.
75	Position Changing Mechanism	Mechanism used for moving a main device from one position to another in an equipment; for example, shifting a removable circuit breaker unit to and from the connected, disconnected, and test positions.	Pneumatic positioner, switchgear elevating mechanism.

Device Number	Device Name	Functional Description	Common GE Application
76	DC Overcurrent Relay	Relay that functions when the current in a dc circuit exceeds a given value.	Magnetic speed “pick-up”, NOx reduction water flow “pick-up.”
77	Telemetering Device	Transmitter used to generate and transmit to a remote location an electrical signal representing a measured quantity, or a receiver used to receive the electrical signal from a remote transmitter and convert the signal to represent the original measured quantity.	
78	Phase-Angle Measuring or Out-of-Step Protective Relay	Relay that functions at a predetermined phase angle between two voltages or between two currents or between voltage and current.	
79	AC Reclosing Relay	Relay that controls the automatic reclosing and locking out of an ac circuit interrupter.	
80	Flow Switch	Switch which operates on given values, or on a given rate of change of flow.	Flow Switch, e.g., cooling air, gas fuel wash water, etc.
81	Frequency Relay	Relay that responds to the frequency of an electrical quantity, operating when the frequency or rate of change of frequency exceeds or is less than a predetermined value.	Generator under-frequency relay.
82	DC Loading Measuring Reclosing Relay	Relay that controls the automatic closing and reclosing of a dc circuit interrupter, generally in response to load circuit conditions.	Automatic circuit restoration relay, field removal relay, line breaker trip relay.
83	Automatic Selective Control or Transfer Relay	Relay that operates to select automatically between certain sources or conditions in equipment, or automatically performs a transfer operation.	Transfer relay, e.g., generator/ synchronous condenser transfer, load sharing voltage control transfer, etc.

Device Number	Device Name	Functional Description	Common GE Application
84	Operating Mechanism	Complete electrical mechanism or servo-mechanism, including the operating motor, solenoids, positions switches, etc. for a tap changer, induction regulator, or any similar piece of apparatus which otherwise has no device function number.	Limit switch, e.g., fuel firing fuel transfer, automatic load application, etc.
85	Carrier or Pilot-Wire Receive Relay	Relay operated or restrained by a signal used in connection with carrier current or dc pilot-wire fault relaying.	
86	Lock-out Relay	Hand reset or electrically reset auxiliary relay operated upon occurrence of abnormal conditions which maintains associated equipment or devices inoperative until it is reset.	Lock-out relay, e.g., atomizing air, bus differential, loss of flame, generator differential, etc.
87	Differential Protective Relay	Protective relay that functions on a percentage or phase angle or other quantitative difference of two currents, or of some other electrical quantities.	Differential protective relay, e.g., bus differential, generator differential, etc.
88	Auxiliary Motor or Motor Generator	Used for operating auxiliary equipment such as pumps, blowers, exciters. rotating magnetic amplifiers, etc.	Auxiliary motors, e.g., lube oil pump-AC, lube oil pump-DC, starting motor, etc.
89	Line Switch	Switch used as a disconnecting, load-interrupter, or isolating switch in ac or dc power. (This device function number is normally not necessary unless the switch is electrically-operated or has electrical accessories such as an auxiliary switch, magnetic lock, etc.).	Auxiliary power, running, disconnect switch.
90	Regulating Device	Device that regulates a quantity or quantities (such as voltage current, power, speed, frequency. temperature, and load) at a certain value or between certain (generally close) limits for machines, tie-lines, or other apparatus.	Regulating devices, e.g., NOx reduction water/steam flow, fuel additive flow control, load regulation voltage regulation.

Device Number	Device Name	Functional Description	Common GE Application
91	Voltage Directional Relay	Relay that operates when the voltage across an open circuit breaker or contactor exceeds a given value in a given direction.	
92	Voltage and Power Directional Relay	Relay that permits or causes the connection of two circuits when the voltage difference between them exceeds a given value in a predetermined direction and causes these two circuits to be disconnected from each other when the power flowing between them exceeds a given value in the opposite direction.	
93	Field Changing Contactor	Contactor that increases or decreases, in one step, the value of field excitation on a machine.	Dead load pickup-exciter field contactor.
94	Tripping or Trip-Free Relay	Relay that trips a circuit breaker, contactor, or equipment, or to permit immediate tripping by other devices; or to prevent immediate reclosure of circuit interrupter if it should open automatically even though its closing circuit is maintained closed.	Trip shut-down relay.
95		Numbers 95 through 99 are used only for specific applications in individual installations if none of the functions assigned to the numbers between 1 and 94 are suitable.	Detector, e.g., compressor surge, metal particle conductivity, humidity, etc.; transient suppressor, e.g., fuel gas pressure compressor discharge pressure, ignition XFMR spark plug, etc.
96			Transducers, e.g., pressure transmitter, LVDT, etc.